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	TITLE : PENGOPERASIAN DAN PERAWATAN TERHADAP SISTEM VENTILASI, PEMANAS, TATA UDARA OPERATION AND MAINTENANCE OF HEATING VENTILATION AIR CONDITIONING SYSTEM	SOP No. : SOP/EBI/EN-016 Revision : 07 Effective Date : Reviewed Date :
PREPARED BY :	REVIEWED BY :	APPROVED BY :
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1. TUJUAN

PURPOSE

- 1.1. Memberikan petunjuk tentang pengoperasian dan perawatan terhadap sistem Ventilasi, Pemanas, Tata Udara dan Pendingin agar sesuai dengan standar yang ada di PT Etana Biotechnologies Indonesia. Yang mencakup:

Provides instruction about operation and maintenance of Heating, Ventilation, Air Conditioning (HVAC) to confirm with standard in PT. Etana Biotechnologies Indonesia. Which include:

- 1.1.1. Unit Penanganan Udara
Air Handling Unit (AHU).
- 1.1.2. Unit Kumparan Kipas
Fan Coil Unit (FCU).
- 1.1.3. Fan Exhaust
Exhaust Fan.
- 1.1.4. Chiller Carrier Evergreen 23XRV.
Carrier Evergreen 23XRV Chiller.
- 1.1.5. Pompa sirkulasi air dingin, air pendingin dan air panas.
Chilled water, cooling water and hot water circulation pumps.
- 1.1.6. Menara pendingin.
Cooling tower.
- 1.1.7. Pembangkit air panas.
Hot water generator.
- 1.1.8. Filter HEPA.
HEPA Filter.

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- 1.2. Menjaga dan memastikan bahwa Sistem HVAC selalu beroperasi dengan normal, menjaga jumlah partikel, suhu, aliran udara, pertukaran udara, tekanan dan kelembaban ruangan sesuai dengan standar yang telah ditetapkan, untuk mendukung semua kegiatan terutama proses produksi di PT Etana Biotechnologies Indonesia.

Maintain and ensure that the HVAC System is always operating normally, maintain the number of particles, temperature, air flow, air changes, pressure and humidity of the room in accordance with established standards, to support all activities, especially the production process at PT Etana Biotechnologies Indonesia.

2. RUANG LINGKUP

SCOPE

SOP ini digunakan oleh departement Engineering khususnya teknisi HVAC sebagai pedoman dalam melakukan pengoperasian dan perawatan terhadap semua Sistem HVAC yang ada di area Produksi, QA, QC, Warehouse & CUB.

This SOP is used by engineering department especially HVAC technician as guidance in order to operation and maintenance of all HVAC systems in the Production, QA, QC, Warehouse & CUB areas.

3. TANGGUNG JAWAB

RESPONSIBILITY

- 3.1. Teknisi HVAC bertanggung jawab :

HVAC technician are responsible :

- 3.1.1. Mengoperasikan sistem HVAC sesuai dengan kebutuhan.

Operate the HVAC system as needed.

- 3.1.2. Melakukan perawatan sesuai jadwal yang telah ditetapkan.

Perform maintenance according to a predefined schedule.

- 3.1.3. Melakukan pemantauan harian terhadap sistem HVAC.

Perform daily monitoring of HVAC system.

- 3.1.4. Menindaklanjuti Permintaan Jasa Engineering (PJE) dari departemen lain yang menyangkut sistem HVAC.

Follow up on Engineering Services Requests from other departments concerning HVAC systems.

- 3.1.5. Melaporkan setiap ada penyimpangan kepada Supervisor Engineering.

Report any deviation to Engineering Supervisor.

- 3.2. Supervisor Engineering bertanggung jawab :

Engineering Supervisor is responsible :

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- 3.2.1. Membuat jadwal perawatan secara berkala.
Make schedule of maintenance periodically.
- 3.2.2. Memastikan pelaksanaan perawatan sesuai dengan jadwal.
Ensure implementation of maintenance is according to the schedule.
- 3.2.3. Memastikan semua alat ukur yang terpasang dan digunakan telah sesuai dengan spesifikasi yang telah ditentukan (dikalibrasi).
Ensure that all measuring instruments installed and used are in accordance with the specifications (calibrated).
- 3.2.4. Memberikan training kepada departemen lain yang terkait.
Provide training to others department which related.
- 3.2.5. Memastikan ketersediaan part kritikal dan jumlahnya cukup.
Ensure availability of critical parts and sufficient quantities.
- 3.2.6. Mengkoordinasikan dan melaporkan kepada Manager Engineering jika terdapat kerusakan pada HVAC System.
Coordinate and report to Engineering Manager if there is damage to HVAC System.
- 3.3. Manajer Engineering bertanggung jawab :
Engineering Manager responsible :
 - 3.3.1. Menentukan spare parts pengganti apabila spare parts yang sesuai sudah tidak dijual lagi.
Determine spare parts if suitable spare parts are no longer sold.
 - 3.3.2. Memutuskan dan mengkoordinir tindakan yang diperlukan, apabila perlu dilakukannya penggantian, modifikasi peralatan, perubahan sistem HVAC.
Decide and coordinate action required, if necessary replacement, equipment modification, change system of HVAC System.

4. DEFINISI

DEFINITION

- 4.3. Sistem Tata Udara adalah suatu sistem yang mengkondisikan lingkungan melalui pengendalian suhu, kelembaban nisbi, arah pergerakan udara dan mutu udara – termasuk pengendalian partikel dan pembuangan kontaminan yang ada di udara (seperti ‘vapors’ dan ‘fumes’). Disebut “sistem” karena AHU terdiri dari beberapa mesin/alat yang masing-masing memiliki fungsi yang berbeda, yang terintegrasi sedemikian rupa sehingga membentuk suatu sistem tata udara yang dapat mengontrol suhu, kelembaban, tekanan udara, tingkat

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kebersihan, pola aliran udara serta jumlah pergantian udara di ruang produksi sesuai dengan persyaratan ruangan yang telah ditentukan.

The Air Conditioning System is a system that conditions the environment through controlling temperature, relative humidity, air movement direction and air quality - including controlling particles and removing contaminants in the air (such as 'vapors' and 'fumes'). Called "system" because AHU consists of several machines/devices which each have different functions, which are integrated in such a way as to form an air system that can control temperature, humidity, air pressure, cleanliness level, airflow pattern and number of air changes in the production room in accordance with the specified room requirements.

- 4.3. Tujuan dari desain Sistem Tata Udara adalah untuk menyediakan sistem sesuai dengan ketentuan CPOB untuk memenuhi kebutuhan perlindungan produk dan proses sejalan dengan persyaratan GEP (*Good Engineering Practices*), seperti keandalan, perawatan, keberlanjutan, fleksibilitas, dan keamanan.

The purpose of the design of the Air Conditioning System is to provide a system in accordance with GMP provisions to meet product and process protection requirements in line with GEP (Good Engineering Practices) requirements, such as reliability, maintenance, sustainability, flexibility and security.

- 4.3. Desain Sistem Tata Udara mempengaruhi tata letak ruang berkaitan dengan hal seperti posisi ruang penyangga udara (*airlock*) dan pintu. Tata letak ruang memberikan efek pada arah aliran udara, perbedaan tekanan udara ruangan dan pengendalian potensi kontaminasi silang. Pencegahan kontaminasi dan kontaminasi silang merupakan suatu pertimbangan desain yang esensial dari sistem Tata Udara. Mengingat aspek kritis ini, desain Sistem Tata Udara harus dipertimbangkan pada tahap desain konsep industri farmasi.

The design of the air-conditioning system influences the layout of the space relating to things such as the position of the airlock and the door. Space layout has an effect on the air flow direction, air pressure differences and control of potential for cross contamination. Prevention of contamination and cross contamination is an essential design consideration of the Air System. In view of this critical aspect, the design of the Air Conditioning System must be considered at the design stage of pharmaceutical industry concept.

5. PETUNJUK UMUM

GENERAL INSTRUCTION

- 5.1. Cara pengoperasian harus dengan benar atau sesuai urutan-urutannya dan perawatan harus dilakukan dengan teratur dan sesuai dengan jadwal yang telah dibuat.

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The way of operation should be correct or in accordance with its sequence and maintenance should be done regularly and in accordance with the schedule that has been made.

- 5.2. Operator/teknisi yang telah terlatih atau sudah mendapatkan pelatihan dan terqualifikasi yang dapat mengoperasikan dan melakukan perawatan sistem HVAC.

Trained or qualified training operator/technicians who can operate and conduct maintenance of HVAC system.

- 5.3. Keselamatan kerja harus tetap di perhatikan selama pengoperasian atau perawatan.

Safety must be kept in mind during the operation or maintenance.

- 5.4. Kegiatan perawatan yang dilakukan secara berkala harus didokumentasikan setelah perawatan dilakukan.

Regular maintenance activities should be documented after maintenance done.

- 5.5. Memastikan alat ukur yang digunakan seperti pressure gauge, thermometer, voltmeter, ampere meter berfungsi dengan baik dan terkalibrasi.

Ensure the measuring instruments used such as pressure gauge, thermometer, voltmeter, ampere meter function properly and calibrated.

- 5.6. Apabila alat ukur tersebut didapati dalam keadaan rusak, harus diberi penandaan yang jelas.

If the measuring device is found to be in a defective state, it should be clearly marked.

- 5.7. Perlengkapan kerja seperti safety dan helm harus digunakan.

Work equipment safety shoes and helmet must used

- 5.8. Setelah melakukan perawatan, semua kegiatan harus di dokumentasikan lengkap pada Formulir Perawatan Berkala terhadap unit penanganan udara No. SOP/EBI/ENG-016-F08.

After conduct maintenance, all activities must be fully documented in the form Periodic Maintenance of Air Handling Unit (AHU) Form No. SOP/EBI/EN-004-F08.

- 5.9. Semua kegiatan perbaikan ataupun perawatan yang dilakukan harus dicatat pada Formulir Kartu Riwayat Mesin No. SOP/EBI/EN/-004-F03.

All repairs or maintenance activities carried out must be recorded on the machine history card Form No. SOP/EBI/EN-004-F03.

- 5.10. Disarankan untuk melakukan pengkondisian HVAC selama 7 jam agar parameter suhu dan RH memenuhi spesifikasi dan memerlukan 35 jam untuk pengkondisian HVAC setelah perawatan AHU jika unit AHU yang mati dan boiler. Tim produksi perlu memastikan bahwa semua parameter (suhu, kelembaban relatif, dan perbedaan tekanan) memenuhi spesifikasi dan melakukan pembersihan ruangan sesuai dengan SOP/EBI/PD-027 versi terbaru sebelum proses produksi.

It is suggested for conditioning HVAC 7 hours for the temperature and RH parameters to meet the required specifications and need 35 hours for conditioning HVAC after maintenance

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AHU if AHU unit off and boiler off. Production team need to ensure that all parameters (temperature, relative humidity and differential pressure) meet specification and perform line clearance according to SOP/EBI/PD-027 latest version before production process.

- 5.11. Setiap suku cadang yang datang dari supplier harus dicek keaslian, dimensi serta kesesuaian spesifikasinya.

Every spare part that comes from the supplier must be checked for authenticity, dimension and suitability of the specification.

- 5.12. Penggantian suku cadang harus sesuai dengan spesifikasi dan fungsinya. Untuk komponen kritis atau suku cadang yang berhubungan langsung dengan produk apabila diganti dengan beda merk dan type yang berbeda tetapi satu jenis maka dibuatkan change control.

Spare parts replacement must be in accordance with the specifications and functions. For critical components or spare parts that are directly related to the product if replaced with different brands and different types but one type, then change control is made.

- 5.13. Untuk perubahan major yang mendasar dan merubah sistem HVAC harus memakai formulir change control dan disetujui oleh manajer terkait.

For major underlying changes and changing the HVAC system must use a change control form and be approved by the relevant manager.

- 5.14. Parameter kritis dari tata udara yang dapat memengaruhi produk adalah :

The critical parameters of the air system that can affect the product are:

- 5.14.1. Suhu.

Temperature.

- 5.14.2. Kelembaban.

Humidity.

- 5.14.3. Partikel Udara (Non Viabel).

Particulate air (non Viabel).

- 5.14.4. Perbedaan tekanan antar ruang dan pola aliran udara.

Pressure Differences between room and airflow patterns.

- 5.14.5. Pertukaran udara.

Air Change.

- 5.14.6. Sistem filtrasi udara.

Air filtration system.

Sistem Tata Udara di PT Etana Biotechnologies Indonesia (EBI) menggunakan Chiller Carrier 23 XRV dengan kapasitas 450 TR yang berjumlah 3 Unit (3x450 TR) yang dioperasikan sesuai kebutuhan, dilengkapi dengan *cooling tower* (3 Unit) serta AHU (20 Unit). Terdapat magnahelic analog yang dapat dilihat langsung pada setiap ruangan yang diharuskan untuk dipantau perbedaannya. Untuk HEPA Filter ruang kelas B, C dan

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D juga terdapat Sistem Monitoring Hepa yang menggunakan magnahelic yang dipantau secara manual pada panel monitoring HEPA. AHU area produksi (AHU B101, AHU B102, AHU B103, AHU B104, AHU B105, B201, B202, B204, B205, B207A, B207B, dan B208) dan AHU area QC (AHU F102, AHU F103, AHU F104, AHU F105, AHU F106, A101 dan A102) dilengkapi dengan system alarm, alarm akan berfungsi apabila supply udara off atau AHU trip sehingga tim engineering akan cepat mengecek AHU.

The Air Conditioning System at PT Etana Biotechnologies Indonesia (EBI) uses a 23 XRV Carrier Chiller with a capacity of 450 TR which amounts to 3 Units (3x450 TR) operated as needed, equipped with Cooling Tower (3 Units) and AHU (20 Units). There is also an analog magnahelic that can be seen directly in each room that is required to monitor the difference in pressure. For HEPA Filter class B, C and D there is also a Hepa Monitoring System that uses magnahelic which is monitored manually on the HEPA Monitoring panel. AHU production area (AHU B101, AHU B102, AHU B103, AHU B104, AHU B105, B201, B202, B204, B205, B207A, B207B, and B208) and AHU QC area (AHU F102, AHU F103, AHU F104, AHU F105, AHU F106, A101 and A102) completed with alarm system, alarm will be turn on if AHU trip or supply air off then engineering team will check the condition off AHU.

6. ALAT DAN BAHAN

TOOL AND MATERIAL

6.1. ALAT

TOOLS

6.1.1. Kain Lap

Duster

6.1.2. Jet cleaner

Jet cleaner

6.1.3. Tang ampere

Ampere meter

6.1.4. Tang

Pliers

6.1.5. Obeng

Screw

6.1.6. Kunci pas

Wrench

6.2. BAHAN

MATERIAL

6.2.1. Air

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Water

6.2.2. *Pelumas / Gemuk*

Lubricant / Grease

7. PROSEDUR

PROCEDURE

7.1. Pengoperasian HVAC.

Operation of HVAC.

7.1.1. Pastikan semua katup oli dan refrigerant sudah pada posisi operasi.

Ensure all oil and refrigerant valves are in the operating position.

7.1.2. Pastikan Air Dingin sudah tersedia dan semua katup Air Dingin sudah pada posisi operasi.

Ensure Chilled Water is available and all Chilled Water valves are in the operating position.

7.1.3. Pastikan Air Dingin sudah tersedia dan semua katup Air Dingin sudah pada posisi operasi.

Ensure Chilled Water is available and all Chilled Water valves are in the operating position.

7.1.4. Pastikan Air Pendingin tersedia dan Menara Pendingin sudah siap beroperasi (Power dan semua katup sudah pada posisi operasi).

Ensure Cooling Water is available and Cooling Tower is ready to operate (Power and all valves are in the operating position).

7.1.5. Pastikan Air Panas tersedia dan Pembangkit Air Panas sudah siap beroperasi (Power dan Plant Steam tersedia serta semua katupnya sudah pada posisi operasi).

Ensure Hot Water is available and Hot Water Generator is ready to operate (Power and Plant Steam are available and all the valves are in the operating position).

7.1.6. Pastikan power pada panel starter/outgoing chiller, menara pendingin, pompa air dingin dan pompa air pendingin sudah ON dan sudah dalam posisi/mode manual 'MAN'.

Ensure the power in the starter/outgoing chiller, cooling tower, chilled water pump and cooling water pump are ON and in the manual position/mode 'MAN'.



Gambar 1. Panel Starter/Outgoing HVAC

Figure 1. Starter Panel/Outgoing HVAC

- 7.1.7. Hidupkan menara pendingin, pompa air dingin dan pompa air pendingin dengan menekan tombol tekan ON (berwarna hijau).
Turn on the CT, CHW and CWP by pressing the push button ON (colored green).
- 7.1.8. Hidupkan Chiller, masuk ke mode lokal dengan menekan tombol 'LOCAL' pada layar panel ICVC Chiller.
Turn on the Chiller, enter local mode by pressing the 'LOCAL' button on the ICVC Chiller screen panel.
- 7.1.9. Hidupkan AHU sesuai dengan area/ruangan yang akan digunakan. Pada panel starter AHU, tekan tombol tekan 'ON' (berwarna hijau).
Turn on the AHU according to the area/room that will be used. On the AHU starter panel, press push button 'ON' (colored green).
- 7.1.10. Hidupkan generator air panas, masuk ke mode manual dan tekan tombol tekan 'ON' (berwarna hijau) pada panel kontrol generator air panas.
Turn on the hot water generator, enter to manual mode and press the push button 'ON' (colored green) on the hot water generator control panel.
- 7.1.11. Khusus untuk AHU QC Micro :
Specifically for AHU of QC Micro : ganti poto



Gambar 2. Panel Starter AHU QC Micro

Figure 2. Starter Panel AHU QC Micro

- 7.1.11.1. Pastikan Saklar Utama sudah hidup.
Ensure the Main Switch is on.
- 7.1.11.2. Pastikan tombol berhenti darurat sudah mati. Jika saklar hidup, putar saklar searah jarum jam.
Ensure the Emergency Stop button is off. If the switch is on, turn the switch clockwise.
- 7.1.11.3. Hidupkan AHU 1, 2, 3, 4, & 5. Tekan tombol kuning satu kali untuk menghidupkan.
Turn on the AHU 1, 2, 3, 4, & 5. Push the yellow button once to turn on.
- 7.1.11.4. Jika AHU sudah hidup, lampu indikator hijau (lihat nomor 4) akan menyala dan nilai pada monitor AHU akan berubah dari 00 ke 40-50.
If the AHU is on, the green light indicator (see number 4) will be on and value in the AHU monitor will change from 00 to 40 – 50.
- 7.1.11.5. Untuk mematikan AHU, pertama tekan tombol kuning satu kali (lihat nomor 4), lampu hijau akan menjadi padam dan nilai pada monitor AHU akan berubah ke 00.
To turn off the AHU, first push the yellow button once (see number 4), the green light will be off and value in AHU monitor will change to 00.
- 7.1.11.6. Untuk mematikan pompa sirkulasi, tekan tombol merah (lihat nomor 3).
To turn off the circulation pump, push the red button (see number 3).

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7.1.12. AHU akan mengatur suhu dan kelembaban ruangan sesuai dengan set point yang sudah ditetapkan sebelumnya.

The AHU will regulate the temperature and humidity of the room according to the set point previously set.

7.1.13. Untuk mematikan HVAC, matikan AHU dengan menekan tombol tekan 'OFF' (berwarna merah) pada panel starter AHU. Kemudian matikan generator air panas dengan menekan tombol tekan 'OFF' (berwarna merah) pada panel kontrol generator air panas. Kemudian matikan Chiller dengan menekan dan menahan tombol 'STOP' selama 1 detik pada layar panel ICVC Chiller. Kemudian matikan menara pendingin, pompa air dingin, pompa air pendingin dengan menekan tombol tekan 'OFF' pada panel starter Chiller.

To turn off HVAC, turn off the AHU by pressing the push button 'OFF' (colored red) on the AHU starter panel. Then turn off the hot water generator by pressing the 'OFF' button (colored red) on the hot water generator control panel. Then turn off the Chiller by pressing and holding the 'STOP' button for 1 second on the ICVC Chiller panel screen. Then turn off the CT, CHWP, CWP by pressing the push button 'OFF' on the Chiller panel starter.

7.2. Pengoperasian terhadap FCU.

Operation of FCU.

7.2.1. Tekan tombol 'ON/OFF' pada remote kontrol.

Press the 'ON/OFF' button on the remote controller.

7.2.2. Atur suhu yang diinginkan.

Set the desired temperature.

7.2.3. Unit FCU akan bekerja mendinginkan ruangan sampai suhu yang diatur.

The FCU unit will work to cool the room to the set temperature.

7.2.4. Untuk mematikan FCU, tekan tombol 'ON/OFF' sekali lagi pada remote kontrol.

To turn off the FCU, press the 'ON/OFF' button again on the remote control.

7.3. Pemantauan harian terhadap sistem HVAC.

Daily monitoring of HVAC System.

7.3.1. Cek suhu *supply* (7 °C) dan *return* (12 °C) air dingin dengan ± 4 °C. Cek pipa *supply* dan *return* chiller terhadap kebocoran dan kondensasi.

Check the temperature of supply (7 °C) and return (12 °C) of chilled water with ± 4 °C.

Check supply and return pipe from leakage and condensation.

7.3.2. Cek suhu *supply* (31 °C) dan *return* (36 °C) air pendingin dengan ± 4 °C.

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Check the temperature of supply (31 °C) and return (36 °C) of cooling water with ± 4 °C.

7.3.3. Cek suhu return (20-55 °C) air panas.

Check the return temperature (20-55 °C) of hot water.

7.3.4. Bersihkan ruangan dan mesin Chiller.

Clean the room and the Chiller machine.

7.3.5. Cek suhu LCL set poin (6-7 °C)

Check the LCL set point temperature

7.3.6. Cek suhu ECL set poin (9-12 °C)

Check the ECL set point temperature

7.3.7. Cek % AMP IN (≥ 40 %)

Check % AMP IN (≥ 40 %)

7.3.8. Cek Oil Level di Sight Glass (≥ 50 %)

Check the Oil Level in the Sight Glass (≥ 50 %)

7.3.9. Cek suhu pada Motor Winding Compressor (≤ 93 °C)

Check the temperature on the Winding Compressor Motor (≤ 93 °C)

7.3.10. Cek suhu pada Discharge Compressed Air (≤ 60 °C)

Check the temperature of the Discharge Compressed Air (≤ 60 °C)

7.3.11. Cek Oil Pressure AP running min 170 kPa

Cek Oil Pressure AP running min 170 kPa

7.3.12. Cek suhu pada Oil Sump (≥ 40 °C)

Check the temperature at the Oil Sump (≥ 40 °C)

7.3.13. Cek Condensor Temperature Liquid In / CDL IN (≥ 28 °C)

Check Condensor Temperature Liquid In / CDL IN (≥ 28 °C)

7.3.14. Cek Condensor Temperature Liquid In / CDL OUT (30-38 °C)

Check Condensor Temperature Liquid In / CDL OUT (30-38 °C)

7.3.15. Cek Evaporator Temperature Liquid In / CHL IN (9-12 °C)

Check Evaporator Temperature Liquid In / CHL IN (9-12 °C)

7.3.16. Cek Evaporator Temperature Liquid Out / CHL OUT (6-7 °C)

Check Evaporator Temperature Liquid Out / CHL OUT (6-7 °C)

7.3.17. Cek suhu pada Evaporator Approach ($\leq 2.8^{\circ}\text{C}$)

Check temperature on Evaporator Approach ($\leq 2.8^{\circ}\text{C}$)

7.3.18. Cek suhu pada Condensor Approach ($\leq 4.0^{\circ}\text{C}$)

Check temperature on Condenser Approach ($\leq 4.0^{\circ}\text{C}$)

7.3.19. Cek suhu pada Condensor Pressure (≤ 1000 kPa)

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Check the temperature at Condenser Pressure (≤ 1000 kPa)

7.3.20. Cek tekanan pada Refrigerant Evaporator (250 kPa - 400 kPa)

Check the pressure on the Refrigerant Evaporator (250 kPa - 400 kPa)

7.3.21. Cek Calc. Evaporator Saturated Temperature (6°C - 7°C)

Check Calc. Evaporator Saturated Temperature (6°C - 7°C)

7.3.22. Cek Evaporator Flow Delta Pressure Cut Out / Min 33.1 kPa (0.33 Bar)

Check Evaporator Flow Delta Pressure Cut Out / Min 33.1 kPa (0.33 Bar)

7.3.23. Cek Condensor Flow Delta Pressure Cut Out / Min 34.5 kPa (0.34 Bar)

Check Condenser Flow Delta Pressure Cut Out / Mini 34.5 kPa (0.34 Bar)

7.3.24. Cek Chiller Watter Pressure IN (3 - 5 Bar)

Check Chilled Water Pressure IN (3 - 5 Bar)

7.3.25. Cek Chiller Watter Pressure OUT (2.5 - 4.5 Bar)

Cek Chiller Watter Pressure OUT (2.5 - 4.5 Bar)

7.3.26. Cek Condensor watter pressure IN (≥ 1.5 Bar)

Check Condenser water pressure IN (≥ 1.5 Bar)

7.3.27. Cek Condensor watter pressure OUT (≥ 1 Bar)

Check Condenser water pressure OUT (≥ 1 Bar)

7.3.28. Cek Refrigerant Pressure/Pressure Gauge (100 psi - 145 psi)

Cek Refrigerant Pressure/Pressure Gauge (100 psi - 145 psi)

7.3.29. Cek Rectifier Temperature ($\leq 71^{\circ}\text{C}$)

Check Rectifier Temperature ($\leq 71^{\circ}\text{C}$)

7.3.30. Cek Inverter Temperature ($\leq 71^{\circ}\text{C}$)

Check Inverter Temperature ($\leq 71^{\circ}\text{C}$)

7.4. Pemantauan Mingguan terhadap Filter setiap hari Senin.

Weekly monitoring of Filter every monday.

7.4.1. Filter Medium F7 (Bag) dengan spesifikasi efisiensi 80-85%.

Medium Filter F7 (Bag) with efficiency specifications 80-85%.

7.4.1.1. Cek perbedaan tekanan filter medium F7 pada setiap AHU. Dengan batas waspada 15 pascal, dan batas tindakan 10 pascal untuk menentukan adanya kebocoran dan batas waspada 335 pascal dan batas tindakan 350 pascal untuk menentukan kemampetan.

Untuk Filter Medium F7 (Bag), Jika angka mencapai batas waspada pada tekanan ≤ 15 pascal dan ≥ 335 pascal, teknisi, melaporkan kepada atasan, melakukan pemeriksaan penyebab sudah mencapai batas waspada dan menyiapkan sparepartnya untuk menjadwalkan penggantian filternya.

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Dan jika angka mencapai batas tindakan pada tekanan ≤ 10 pascal dan ≥ 350 pascal teknisi melaporkan kepada atasan dan kordinasi dengan user untuk menjadwalkan penggantian filter.

Check the F7 medium filter pressure difference on each AHU. With an alert limit of 15 pascals, and an action limit of 10 pascals to determine the presence of leaks and an alert limit of 335 pascals and an action limit of 350 pascals to determine congestion.

For F7 Medium Filter (Bag), If the number reaches the alert limit at pressures of ≤ 15 pascals and ≥ 335 pascals, the technician, reporting to superiors, conducts an examination of the cause of the alert limit being reached and prepares spare parts to schedule a filter replacement.

And if the number reaches the action limit at pressures of ≤ 10 pascals and ≥ 350 pascals, the technician reports to the supervisor and coordinates with the user to schedule a filter replacement.

7.4.1.2. Lakukan penggantian filter medium F7 jika bocor atau mampet.

Replace the medium filter if it is leaking or clogged.

7.4.2. Filter Medium F7 (Rigid Bag) dengan spesifikasi efisiensi 80 – 90 %.

Medium Filter F7 (Rigid Bag) with efficiency specifications 80 – 90 %.

Cek perbedaan tekanan filter medium F7 pada setiap AHU. Dengan batas waspada 15 pascal, dan batas tindakan 10 pascal untuk menentukan adanya kebocoran dan batas waspada 335 pascal dan batas tindakan 350 pascal untuk menentukan kemampetan.

Untuk Filter Medium F7 (Rigid Bag), Jika angka mencapai batas waspada pada tekanan ≤ 15 pascal dan ≥ 335 pascal, teknisi, melaporkan kepada atasan, melakukan pemeriksaan penyebab sudah mencapai batas waspada dan menyiapkan sparepartnya untuk menjadwalkan penggantian filternya.

Dan jika angka mencapai batas tindakan pada tekanan ≤ 10 pascal dan ≥ 350 pascal teknisi melaporkan kepada atasan dan kordinasi dengan user untuk menjadwalkan penggantian filter.

Check the F7 medium filter pressure difference on each AHU. With an alert limit of 15 pascals, and an action limit of 10 pascals to determine the presence of leaks and an alert limit of 335 pascals and an action limit of 350 pascals to determine congestion.

For F7 Medium Filter (Rigid Bag), If the number reaches the alert limit at pressures of ≤ 15 pascals and ≥ 335 pascals, the technician, reporting to superiors, conducts an examination of the cause of the alert limit being reached and prepares spare parts to schedule a filter replacement.

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And if the number reaches the action limit at pressures of ≤ 10 pascals and ≥ 350 pascals, the technician reports to the supervisor and coordinates with the user to schedule a filter replacement.

7.4.3. Filter HEPA H10 dengan spesifikasi efisiensi 85 %.

HEPA Filter H10 with efficiency specifications 85 %.

7.4.3.1. Cek perbedaan tekanan filter HEPA H10 pada setiap AHU. Dengan batas waspada 100 pascal, dan batas tindakan 90 pascal untuk menentukan adanya kebocoran dan batas waspada 485 pascal dan batas tindakan 500 pascal untuk menentukan kemampetan.

Filter HEPA H10, Jika angka mencapai batas waspada pada tekanan ≤ 100 pascal dan ≥ 485 pascal, teknisi, melaporkan kepada atasan, melakukan pemeriksaan penyebab sudah mencapai batas waspada dan menyiapkan sparepartnya untuk menjadwalkan penggantian filternya.

Dan jika angka mencapai batas tindakan pada tekanan ≤ 90 pascal dan ≥ 500 pascal teknisi melaporkan kepada atasan dan kordinasi dengan user untuk menjadwalkan penggantian filter.

Check the HEPA H10 filter pressure difference on each AHU. With an alert limit of 100 pascals, and an action limit of 90 pascals to determine the presence of leaks and an alert limit of 485 pascals and an action limit of 500 pascals to determine congestion.

H10 HEPA filter, If the number reaches the alert limit at pressures of 100 pascals and 485 pascals, the technician, reports to superiors, carries out an examination of the cause of having reached the alert limit and prepares spare parts to schedule filter replacement.

And if the number reaches the action limit at pressures of 90 pascals and 500 pascals, the technician reports to the supervisor and coordinates with the user to schedule a filter replacement.

7.4.3.2. Lakukan penggantian HEPA atau seal jika bocor atau mampet.

Replace the HEPA or seal if it is leaking or clogged.

7.4.4. Filter HEPA H14 dengan spesifikasi efisiensi 99,995 %.

HEPA Filter H14 with efficiency specifications 99,995 %.

7.4.4.1. Cek tekanan absolut Filter Hepa pada panel Hepa Monitoring di Produksi, QA/QC, dan Warehouse sesuai dengan area/bagian yang telah ditetapkan. Dengan batas waspada 100 pascal, dan batas tindakan 90 pascal untuk menentukan adanya kebocoran Hepa, seal hepa atau selang magnahelic.

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Dan batas waspada 485 pascal dan batas tindakan 500 pascal untuk menentukan kemampetan.

Filter HEPA H14, Jika angka mencapai batas waspada pada tekanan ≤ 100 pascal dan ≥ 485 pascal, teknisi, melaporkan kepada atasan, melakukan pemeriksaan penyebab sudah mencapai batas waspada dan menyiapkan sparepartnya untuk menjadwalkan penggantian filternya.

Dan jika angka mencapai batas tindakan pada tekanan ≤ 90 pascal dan ≥ 500 pascal teknisi melaporkan kepada atasan dan kordinasi dengan user untuk menjadwalkan penggantian filter.

Check the absolute pressure of the Hepa Filter on the Hepa Monitoring panel in Production, QA/QC, and Warehouse according to the designated area/section. With an alert limit of 100 pascals, and an action limit of 90 pascals to determine the presence of Hepa leaks, hepa seals or magnahelic tubing. And the alert limit is 485 pascals and the action limit is 500 pascals to determine stagnation.

H14 HEPA filter, if the number reaches the alert limit at a pressure of 100 pascals

and 485 pascals, the technician, reports to superiors, inspects the cause of the

alert limit and prepares spare parts to schedule filter replacement.

And if the number reaches the action limit at pressures of 90 pascals and 500 pascals, the technician reports to the supervisor and coordinates with the user to schedule a filter replacement.

7.4.4.2. Lakukan penggantian HEPA atau seal jika bocor atau mampet.

Replace the HEPA or seal if it is leaking or clogged.

7.5. Perawatan berkala Pre Filter pada setiap AHU dilakukan setiap 1 bulan sekali, dengan cara inspeksi secara visual atau dibersihkan, dan ganti jika filter sudah terlihat kotor. Fresh air filter diganti jika sudah terlihat kotor. Indikator kotornya adalah debu tebal menempel pada filter, Metode pembersihannya adalah dengan di basuh oleh air bertekanan.

Pre-filter periodic maintenance on each AHU is carried out every 1 month, by inspection visually or cleaning, and change the filter if the filter looks dirty. Fresh air filter change if looks dirty. The dirty indicator is thick dust attached to the filter, the cleaning method is to be washed by pressurized water.

7.6. Perawatan berkala 1 bulanan (L1), 3 bulanan (L2), dan 1 tahunan (L4) terhadap Unit Penanganan Udara (AHU).

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Periodic maintenance of 1 month (L1), 3 months (L2), and 1 year (L4) of Air Handling Unit (AHU).

7.6.1. Perawatan 1 bulanan (L1).

1 month maintenance (L1).

7.6.1.1. Cek kondisi basin AHU basah atau kering jika basah buka katup pembuangan

Check the condition of the AHU basin, whether it is wet or dry, if it is wet, open the drain valve

7.6.1.2. Cek kondisi prefilter G4 ganti jika sudah kotor. Jika AHU dilengkapi dengan precooling dan dehumidifier maka bersihkan filternya, pre filter (fresh air) diganti seminggu 1x setiap hari senin, diganti dengan filter yang sudah di cuci (maksimal pencucian filter yg boleh dipakai adalah 10x, setelah di cuci 10x maka filter harus diganti yang baru).

Check the condition of the AHU basin, whether it is wet or dry, if it is wet, open the drain valve and check the condition of the G4 prefilter, replace it if it is dirty. If the AHU is equipped with precooling and dehumidifier then clean the filter, replace the pre filter (fresh air) once a week every Monday, replace it with a filter that has been washed (the maximum filter wash that can be used is 10x, after washing it 10x then the filter must be replaced. new).

7.6.1.3. Cek kondisi insulasi pipa drain AHU.

Check the insulation condition of the AHU drain pipe

7.6.2. Perawatan 3 bulanan (L2).

3 month maintenance (L2).

7.6.2.1. Bersihkan cooling coil & heating coil dengan menggunakan air bertekanan.

Clean the cooling coil & heating coil using pressurized water.

7.6.2.2. Cek tampungan kondensasi basin dan pipa pembuangan, jika mampet tekan dengan air bertekanan atau vakum hingga lancar.

Check the condensation basin and drain pipe, if clogged press with pressurized water or vacuum until smooth.

7.6.2.3. Cek motor blower AHU, Precooling dan dehumidifier dari suara tidak normal, perbaiki atau ganti bila perlu.

Check the blower motor AHU, precooling and dehumidifier for abnormal sound, repair or replace if necessary.

7.6.2.4. Cek V-Belt Motor AHU, precooling dan dehumidifier, setting jika kendur dan ganti jika putus.

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Check Motor V-Belt AHU, precooling and dehumidifier, setting if loose and replace if broken.

- 7.6.2.5. Cek arus listrik, bandingkan dengan spesifikasi pada plat nama, jika lebih dari 10% lakukan investigasi dan perbaikan, bersihkan panel listrik dan filter panel

Check the electric current, compare with the specifications on the name plate, if more than 10% do an investigation and repair, cleaning electrical panel and filter of panel.

- 7.6.3. Perawatan 1 tahunan (L4).

1 year maintenance (L4).

- 7.6.3.1. Cek filter medium (F7) rigid, bag filter dan H10 hepa filter, pergantian dilakukan jika differential pressure sudah mencapai action limit (*blok atau bocor*)

Check medium filter (F7) rigid, bag filter and H10 hepa filter. Change filter if differential pressure hit action limit (blocking or leak)

- 7.6.3.2. Bersihkan tampungan kondensasi dan pipa pembuangan, tekan dengan air bertekanan.

Clean the condensation reservoir and drain pipe, press with pressurized water.

- 7.6.3.3. Ukur tahanan isolasi dan pentanahan, maksimal 5 Ohm.

Measure the isolation resistance and grounding, maximum 5 ohm.

- 7.6.3.4. Periksa dan kencangkan semua koneksi terminal kabel.

Inspect and tightened all cable terminals connections.

- 7.6.3.5. Cek sambungan-sambungan ducting dari kebocoran.

Check ducting connections from leaks.

- 7.6.3.6. Periksa semua breaker (CB) dan komponen listrik lainnya dari perubahan warna disebabkan oleh beban yang berlebihan. Ganti komponen yang usang.

Inspect all circuit breakers and other electric component for discolouration caused by excessive load. Replace worn-out components.

- 7.6.3.7. Cek HEPA Filter H14 dilakukan secara berkala, penggantian setiap 3 tahun sekali atau jika differential pressure masih memenuhi syarat yaitu 90 – 500 Pascal artinya filter masih dapat digunakan.

Check of HEPA filter H14 is done periodically. Chs every 3 years or if the differential pressure is still eligible 90 – 500 Pascal means the filter can still be used.

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7.7. Perawatan berkala 1 bulanan (L1) dan 3 bulanan (L2) terhadap Unit Komponen Kipas (FCU).
Periodic maintenance of 1 month (L1) and 3 months (L2) of the Fan Component Unit (FCU).

7.7.1. Perawatan 1 bulanan (L1).

1 month maintenance (L1).

7.7.1.1. Bersihkan filter fresh air dengan air bertekanan kemudian keringkan dan pasang kembali, ganti jika sudah rusak/sobek.

Clean the fresh air filter with pressurized water then dry and reinstall it, replace it if it has been damaged/torn.

7.7.1.2. Cek coil pendingin, bersihkan jika kotor menggunakan air bertekanan.

Check the cooling coil, clean if dirty using pressurized water.

7.7.2. Perawatan 3 bulanan (L2).

3 month maintenance (L2).

7.7.2.1. Bersihkan coil pendingin menggunakan air bertekanan.

Clean the cooling coil using pressurized water.

7.7.2.2. Cek tampungan kondensasi basin dan pipa pembuangan, jika mampet tekan dengan air bertekanan atau vakum hingga lancar.

Check the condensation basin and drain pipe, if clogged press with pressurized water or vacuum until smooth.

7.7.2.3. Cek fan motor dari suara tidak normal, perbaiki atau ganti jika perlu.

Check motor fan from abnormal sound, repair or replace if necessary.

7.7.2.4. Cek sambungan koneksi listrik, kencangkan jika kendur.

Check the connection of the electrical connection, tighten if loose.

7.7.2.5. Cek arus listrik, bandingkan dengan spesifikasi pada plat nama, jika lebih dari 10% lakukan investigasi dan perbaikan.

Check the electric current, compare with the specifications on the name plate, if more than 10% do an investigation and repair.

7.8. Perawatan berkala 3 bulanan (L2) dan 1 Tahunan (L4) terhadap Exhaust Fan (EF).
Periodic maintenance of 3 months (L2) and 1 year (L4) of Exhaust Fan (EF).

7.8.1. Perawatan 3 bulanan (L2).

3 month maintenance (L2).

7.8.1.1. Cek motor blower dari suara tidak normal, perbaiki atau ganti bila perlu.

Check the blower motor for abnormal sound, repair or replace if necessary.

7.8.1.2. Cek arus listrik, bandingkan dengan spesifikasi pada plat nama, jika lebih dari 10% lakukan investigasi dan perbaikan.

Check the electric current, compare with the specifications on the name plate, if more than 10% do an investigation and repair.

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7.8.1.3. Cek V-Belt Motor, setting jika kendur dan ganti jika putus.

Check Motor V-Belt, setting if loose and replace if broken.

7.8.1.4 Cek filter karbon atau bag filter F6, ganti jika mampet atau bocor.

Check the carbon filter or F6 filter bag, replace if it is clogged or leaking.

7.9. Perawatan 1 tahunan (L4).

1 year maintenance (L4).

7.9.1. Cek filter karbon atau bag filter F6 dan filter bibo, pergantian dilakukan jika differential pressure sudah mencaapai action limit atau kotor (blocking)

Check the carbon filter or bag filter F6 and bibo filter, replacement is carried out if the differential pressure has reached the action limit or is dirty (blocking)

7.9.2. Ukur tahanan isolasi dan pentanahan, maksimal 2 Ohm.

Measure the isolation resistance and grounding, maximum 2 ohm.

7.9.3. Cek sambungan-sambungan ducting dari kebocoran.

Check ducting connections from leaks.

7.9.4. Periksa semua breaker (CB) dan komponen listrik lainnya dari perubahan warna disebabkan oleh beban yang berlebihan. Ganti komponen yang usang.

Inspect all circuit breakers and other electric component for discolouration caused by excessive load. Replace worn-out components.

7.10. Perawatan berkala 3 bulanan (L2) dan 1 bulanan (L4) terhadap Chiller Carrier Evergreen 23XRV.

Periodic maintenance of 3 months (L2) and 1 month (L4) of Chiller Carrier Evergreen 23XRV.

7.10.1. Perawatan 3 Bulanan (L2).

3 month maintenance (L2).

7.10.1.1. Cek level Oli kompressor pada sightglass saat kompressor running, jika kurang dari 50% tambahkan oli.

Check compressor oil level on sight glass when compressor is running, if less than 50% add oil.

7.10.1.2. Bersihkan panel kontrol dari debu menggunakan vakum dan kencangkan koneksi kabel jika kendur. Pastikan sebelumnya power listrik ke panel kontrol sudah off.

Clean the control panel from dust using a vacuum and tighten the cable connection if it is loose. Make sure the power supply to the control panel is off.

7.10.1.3. Periksa pompa dari kebocoran, suara kasar atau malfungsi lainnya. Perbaiki atau ganti jika perlu.

Inspect the pumps from leaks, rough noise or other malfunction. Repair or

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replace if necessary.

7.10.1.4. Cek sambungan koneksi listrik, kencangkan jika kendur.

Check the connection of the electrical connection, tighten if loose.

7.10.1.5. Cek arus listrik pompa, bandingkan dengan spesifikasi pada plat nama, jika lebih dari 10% lakukan investigasi dan perbaikan.

7.10.1.6. *Check the electric current of the pump, compare with the specifications on the name plate, if more than 10% do an investigation and repair.*

7.10.1.7. Bersihkan strainer.

Clean the strainer.

7.10.2. Perawatan 1 tahunan (L4).

1 year maintenance (L4).

7.10.2.1. Ganti Oli Kompresor.

Change Compressor Oil.

7.10.2.2. Ganti Filter Oli.

Change Oil Filter.

7.10.2.3. Ganti filter pengering pendingin motor.

Change motor cooling filter-drier.

7.10.2.4. Periksa dan bersihkan tabung pendingin.

Inspect and clean cooler tubes.

7.10.2.5. Periksa fungsi katup relief pendingin.

Inspect function of the cooler relief valves.

7.10.2.6. Ganti filter pengering refrigerant.

Change refrigerant filter-drier.

7.10.2.7. Periksa dan bersihkan tabung kondensor.

Inspect and clean condensor tubes.

7.10.2.8. Periksa fungsi katup relief kondensor.

Inspect function of the condensor relief valves.

7.10.2.9. Ganti VFD strainer refrigerant.

Change VFD refrigerant strainer.

7.10.2.10. Ukur tahanan isolasi dan pentanahan, maximal 5 Ohm.

Measure the isolation resistance and grounding, maximum 5 ohm.

7.10.2.11. Periksa dan kencangkan semua koneksi terminal kabel.

Inspect and tightened all cable terminals connections.

7.10.2.12. Cek mekanikal sil pompa, ganti jika terlihat rusak atau bocor.

Check the mechanical pump seal, replace if it looks damaged or leaks.

7.10.2.13. Periksa semua circuit breaker (CB) dan komponen listrik lainnya dari

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perubahan warna disebabkan oleh beban yang berlebihan.

Ganti komponen yang usang.

Inspect all circuit breakers and other electric component for discolouration caused by excessive load. Replace worn-out components.

7.11. Perawatan berkala 3 bulanan (L2) dan 1 tahunan (L4) terhadap Tower Pendingin.

Periodic maintenance of 3 months (L2) and 1 year (L4) of Cooling Tower.

7.11.1. Perawatan 3 Bulanan (L2).

3 month maintenance (L2).

7.11.1.1. Periksa sambungan-sambungan pipa, dan katup dari kebocoran.

Inspect the pipe connections and valves for leaks.

7.11.1.2. Cek fan blower dari suara kasar atau malfungsi lainnya. Perbaiki atau ganti jika perlu.

Check blower fan from rough noise or other malfunctions. Repair or replace if necessary.

7.11.1.3. Bersihkan sirip-sirip dan bak penampungan air dari kotoran yang mungkin menempel seperti plastik dll.

Clean the fins and water reservoirs from dirt that might stick like plastic etc.

7.11.1.4. Periksa pompa dari kebocoran, suara kasar atau malfungsi lainnya.

Perbaiki atau ganti jika perlu.

Inspect the pumps from leaks, rough noise or other malfunction. Repair or replace if necessary.

7.11.1.5. Cek sambungan koneksi listrik, kencangkan jika kendur.

Check the connection of the electrical connection, tighten if loose.

7.11.1.6. Cek arus listrik motor blower & pompa, bandingkan dengan spesifikasi pada plat nama, jika lebih dari 10% lakukan investigasi dan perbaikan.

Check the electric current of the blower motor & pumps, compare with the specifications on the name plate, if more than 10% do an investigation and repair.

7.11.1.7. Bersihkan strainer.

Clean the strainer.

7.11.2. Perawatan 1 tahunan (L4).

1 year maintenance (L4).

7.11.2.1. Ukur tahanan isolasi dan pentanahan, maximal 5 Ohm.

Measure the isolation resistance and grounding, maximum 5 ohm.

7.11.2.2. Periksa dan kencangkan semua koneksi terminal kabel.

Inspect and tightened all cable terminals connections.

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7.11.2.3. Cek mekanikal sil pompa, ganti jika terlihat rusak atau bocor.

Check the mechanical pump seal, replace if it looks damaged or leaks.

7.11.2.4. Periksa semua breaker (CB) dan komponen listrik lainnya dari perubahan warna disebabkan oleh beban yang berlebihan. Ganti komponen yang usang.

Inspect all circuit breakers and other electric component for discolouration caused by excessive load. Replace worn-out components.

7.12. Perawatan berkala 3 bulanan (L2) dan 1 bulanan (L4) terhadap Pembangkit Air Panas.

Periodic maintenance of 3 months (L2) and 1 month (L4) of Hot Water Generator.

7.12.1. Perawatan 3 bulanan (L2).

3 month maintenance (L2).

7.12.1.1. Bersihkan panel kontrol dari debu menggunakan vakum dan kencangkan koneksi kabel jika kendur. Pastikan sebelumnya power listrik ke panel kontrol sudah off.

Clean the control panel from dust using a vacuum and tighten the cable connection if it is loose. Make sure the power supply to the control panel is off.

7.12.1.2. Periksa pompa dari kebocoran, suara kasar atau malfungsi lainnya. Perbaiki atau ganti jika perlu.

Inspect the pumps from leaks, rough noise or other malfunction. Repair or replace if necessary.

7.12.1.3. Cek sambungan koneksi listrik, kencangkan jika kendur.

Check the connection of the electrical connection, tighten if loose.

7.12.1.4. Cek arus listrik pompa, bandingkan dengan spesifikasi pada plat nama, jika lebih dari 10% lakukan investigasi dan perbaikan.

Check the electric current of the pump, compare with the specifications on the name plate, if more than 10% do an investigation and repair.

7.12.1.5. Cek fungsi katup pengaman. Perbaiki atau ganti jika perlu.

Check the function of the safety valve. Repair or replace if necessary.

7.12.1.6. Bersihkan strainer.

Clean the strainer.

7.12.2. Perawatan 1 tahunan (L4).

1 year maintenance (L4).

7.12.2.1. Periksa semua steam trap dari kebocoran atau kerusakan lainnya, perbaiki atau ganti jika perlu.

Inspect all steam traps from leaks or other malfunction, repair or replace if necessary.

7.12.2.2. Cek mekanikal sil pompa, ganti jika terlihat rusak atau bocor.

Check the mechanical pump seal, replace if it looks damaged or leaks.

7.12.2.3. Bongkar dan bersihkan Katup Pengurang Tekanan (PRV) dan pasang kembali dengan menggunakan suku cadang yang termasuk dalam kit servis katup.

Disassemble and clean the Pressure Reducing Valves (PRV) and rebuild it by using spare parts included in the valve service kit

7.13. Daftar suku cadang kritikal yang berdampak langsung ke kualitas :

List of critical parts that have a direct impact on quality :

7.13.1. Filter HEPA H14.

HEPA Filter H14.

7.13.2. Filter HEPA H10.

HEPA Filter H10.

7.13.3. Filter Medium F7 (Rigid Bag).

Medium Filter F7 (Rigid Bag).

7.13.4. Magnehelic.

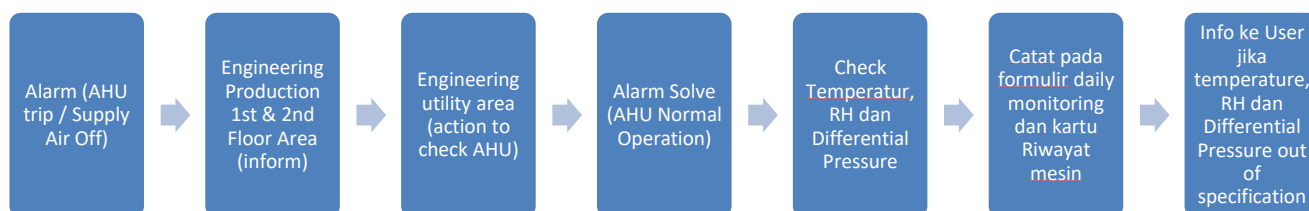
Magnehelic.

7.14. Jika alarm menyala ikuti prosedur dibawah ini:

If alarm turn on follow procedure below:

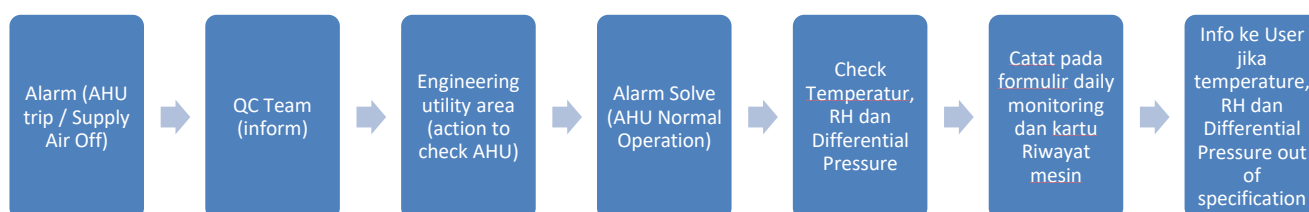
7.14.1. Alarm AHU area produksi:

Alarm AHU production area



7.14.2. Alarm AHU area QC

Alarm AHU QC Area



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8. REFERENSI

REFERENCE

- 8.1. *Start-Up, Operation and Maintenance Instructions of Carrier Evergreen 23XRV Chiller.*
- 8.2. *SDF Operational Service and Maintenance.*
- 8.3. Petunjuk Teknis Sarana Penunjang Kritis Industri Farmasi.

9. LAMPIRAN

ATTACHMENT

- 9.1. Logbook Pemantauan Harian terhadap Sistem HVAC Chiller dan Hot water No. SOP/EBI/EN-016-F01H.
Daily Monitoring of HVAC, Chiller and hot water System Logbook No. SOP/EBI/EN-016-F01H.
- 9.2. Formulir Pemantauan Mingguan terhadap Filter Medium F7 Bag, F7 Rigid, dan H10 No. SOP/EBI/EN-016-F03H.
Weekly Monitoring of Medium Filter F7 (Bag), rigid bag and H10 Form No. SOP/EBI/EN-016-F03H.
- 9.3. Formulir Pemantauan Mingguan terhadap Filter HEPA H14 No. SOP/EBI/EN-016-F06H.
Weekly Monitoring of HEPA Filter Form No. SOP/EBI/EN-016-F06H.
- 9.4. Formulir Pembersihan Fresh Air Filter dan Pre Filter G4 AHU No. SOP/EBI/EN-016-F07H.
Cleaning of AHU Fresh Air Filter and Pre Filter G4 No. SOP/EBI/EN-016-F07H.
- 9.5. Formulir Perawatan Berkala terhadap Unit Penanganan Udara No. SOP/EBI/EN-016-F08H.
Periodic Maintenance of Air Handling Unit Form No. SOP/EBI/EN-016-F08H.
- 9.6. Formulir Perawatan Berkala terhadap Unit Kumpanan Kipas No. SOP/EBI/EN-016-F09H.
Periodic Maintenance of Fan Coil Unit Form No. SOP/EBI/EN-016-F09H.
- 9.7. Formulir Perawatan Berkala terhadap Fan Exhaust No. SOP/EBI/EN-016-F10H.
Periodic Maintenance of Exhaust Fan Form No. SOP/EBI/EN-016-F10H.
- 9.8. Formulir Perawatan Berkala terhadap Chiller Chiller Carrier Evergreen 23XRV No. SOP/EBI/EN-016-F11H.
Periodic Monitoring of Chiller Chiller Carrier Evergreen 23XRV No. SOP/EBI/EN-016-F11H.
- 9.9. Formulir Perawatan Berkala terhadap Menara Pendingin No. SOP/EBI/EN-016-F12H.
Periodic Monitoring of Cooling Tower No. SOP/EBI/EN-016-F12H.
- 9.10. Formulir Perawatan Berkala terhadap Generator Air Panas No. SOP/EBI/EN-016-F13H.
Periodic Maintenance of Hot Water Generator Form No. SOP/EBI/EN-016-F13H.

10. RIWAYAT HISTORY

Tanggal <i>Date</i>	No. Change Control <i>Change Control No.</i>	No. Revisi <i>No. Revision</i>	Alasan Perubahan <i>Reason of Change</i>
17 Jun 2019	N/A	00	Dokumen baru <i>New Document</i>
07 Aug 2019	N/A	01	Penambahahan prosedur pengoperasian dan perawatan terhadap HVAC gedung produksi, warehouse dan CUB. <i>Added operating and maintenance procedures for HVAC production buildings, warehouses and CUB.</i>
16 Oct 2019	N/A	02	1. Penambahan batas waspada dan batas tindakan terhadap perbedaan tekanan Medium Filter F7, HEPA filter H10 pada AHU. <i>Added alert limits and action limits for pressure differences in Medium Filter F7, HEPA filter H10 at AHU.</i> 2. Penambahan batas waspada dan batas tindakan terhadap HEPA Monitoring H14 pada poin 6.4.4.1. <i>Added alert limits and action limits on HEPA Monitoring H14 points 6.4.4.1.</i>

10 Dec 2021	CC-EBI-EN-013-21	03	1. Memberikan penjelasan spesifikasi filter untuk medium filter dan HEPA filter. <i>Provide an explanation of filter specifications for medium filters and HEPA filters.</i> 2. Memberikan penjelasan yang harus dilakukan jika mencapai batas waspada dan batas Tindakan. <i>Provide an explanation of what to do if it reaches the alert limit and the Action limit.</i> 3. Menentukan untuk penggantian HEPA Filter H14 dilakukan secara berkala setiap 3 tahun sekali. <i>Determine to replace the HEPA Filter H14 periodically every 3 years.</i>
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23 FEB 2022	CC-EBI-EN-001-22	04	1. Revisi point no. 5 (petunjuk umum), menambahkan informasi alarm AHU. <i>Revision point Number 5 (General Instruction) add information alarm AHU.</i> 2. Revisi point. No. 6 (prosedur) Menambahkan alur atau flow diagram Ketika terjadi alarm dengan mencantumkan Tindakan dari sisi teknis dan dokumentasi pada formulir daily monitoring HVAC system. <i>Revision point no. 6 (procedure) add flow diagram if alarm rise and list technical and documentation action on formulir daily monitoring HVAC system.</i>
	CC-EBI-EN-014-22	05	1. Menambahkan pemeriksaan kondisi basin AHU (basah/ kering) setiap bulan pada point 7.6.1. <i>Add checking condition of basin AHU (Wet / dry) Every month on point 7.6.1.</i> 2. Menghilangkan point 3.2.4 karena review dilakukan setiap hari oleh supervisor engineering. <i>Delete point 3.2.4 because engineering supervisor review every day.</i>

			3. Menambahkan sepatu safety pada poin 5.7 sebagai peralatan safety yang harus digunakan. <i>Add safety shoes on point 5.7 as safety equipment.</i> 4. Menambahkan nomor formulir yang digunakan pada point 5.9 <i>Add no form on point 5.9</i> 5. Menambahkan pemeriksaan filter precooling, dehumidifier, motor fan precooling, dehumidifier, v-belt precooling, dehumidifier, Rotor drive motor assembly, Dehumidification rotor, dan Dehumidification burner pada perawatan L1, L2 dan L4 pada poin 7.6 <i>Add Checking filter precooling, dehumidifier, motor fan precooling, dehumidifier, v-belt precooling, dehumidifier, Rotor drive motor assembly, Dehumidification rotor, and Dehumidification burner on maintenance L1, L2 dan L4 on poin 7.6</i> 6. Menggabungkan form pemantauan HVAC menjadi Pemantauan HVAC, Chiller dan Hot water No. SOP/EBI/EN-016-F01 <i>Merge form of monitoring HVAC become Monitoring HVAC, Chiller, and Hot water No. SOP/EBI/EN-016-F01.</i>
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			7. Menambahkan penggantian filter BIBO pada exhaust fan perawatan level 4 pada poin 7.9.1. <i>Add change BIBO Filter on exhaust fan maintenance L4 on point 7.9.1.</i> 8. Menggabungkan formulir pemantauan filter F7 bag filter, rigid bag dan filter H10 pada poin 9.2. <i>Merge form monitoring of filter F7 bag filter, rigid bag and H10 on point 9.2.</i> 9. Menghilangkan point 3.3 terkait dengan review kinerja operator HVAC karena tidak relevan dengan SOP ini. <i>Delete point 3.3 because the point is not relevant to this SOP.</i> 10. Merevisi point 7.4 menjadi pemantauan mingguan setiap hari Senin. <i>Revised point 7.4 to weekly monitoring every monday.</i> 11. Menambahkan indikator kotor pada filter, pada poin 7.6.1 dan poin 7.5 <i>Add dirty indicator on filter and method to clean on point 7.6.1 and 7.5</i>
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31 Jan 2024	CC-EBI-QA-013-24	06	- Menambahkan Cek suhu LCL set poin (6-7 °C) pada poin 7.3.5. <i>Added LCL temperature check set point (6-7 °C) in point 7.3.5..</i> - Menambahkan Cek suhu ECL set poin (9-12 °C) pada poin 7.3.6. <i>Added ECL temperature check set point (9-12 °C) in point 7.3.6.</i> - Menambahkan Cek % AMP IN (≥ 40 %) pada poin 7.3.7. <i>Add Check % AMP IN (≥ 40 %) on point 7.3.7.</i> - Menambahkan Cek Oil Level di Sight Glass (≥ 50 %) pada poin 7.3.8. <i>Add Check the Oil Level in the Sight Glass (≥ 50 %) on point 7.3.8.</i> - Menambahkan Cek suhu pada Motor Winding Compressor (≤ 93 °C) pada poin 7.3.9. <i>Add Check the temperature on the Winding Compressor Motor (≤ 93 °C) on point 7.3.9.</i> - Menambahkan Cek suhu pada Discharge Compressed Air (≤ 60 °C) pada poin 7.3.10. <i>Add Check the temperature of the Discharge Compressed Air (≤ 60 °C) on point 7.3.10.</i>
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7. Menambahkan Cek Oil Pressure AP running min 170 kPa pada poin 7.3.11.
Add Cek Oil Pressure AP running min 170 kPa on point 7.3.11.
8. Menambahkan Cek suhu pada Oil Sump ($\geq 40\text{ }^{\circ}\text{C}$) pada poin 7.3.12.
Add Check the temperature at the Oil Sump ($\geq 40\text{ }^{\circ}\text{C}$) on point 7.3.12.
9. Menambahkan Cek Condensor Temperature Liquid In / CDL IN ($\geq 28\text{ }^{\circ}\text{C}$) pada poin 7.3.13.
Add Check Condensor Temperature Liquid In / CDL IN ($\geq 28\text{ }^{\circ}\text{C}$) on point 7.3.13.
10. Menambahkan Cek Condensor Temperature Liquid In / CDL OUT (30-38 $^{\circ}\text{C}$) pada poin 7.3.14.
Add Check Condensor Temperature Liquid In / CDL OUT (30-38 $^{\circ}\text{C}$) on point 7.3.14.
11. Menambahkan Cek Evaporator Temperature Liquid In / CHL IN (9-12 $^{\circ}\text{C}$) pada poin 7.3.15.
Add Check Evaporator Temperature Liquid In / CHL IN (9-12 $^{\circ}\text{C}$) on point 7.3.15.

12. Menambahkan Cek Evaporator Temperature Liquid Out / CHL OUT (6-7 °C) pada poin 7.3.16.
Add Check Evaporator Temperature Liquid Out / CHL OUT (6-7 °C) on point 7.3.16.
13. Menambahkan Cek suhu pada Evaporator Approach ($\leq 2.8^{\circ}\text{C}$) pada poin 7.3.17.
Add Check temperature on Evaporator Approach ($\leq 2.8^{\circ}\text{C}$) on point 7.3.17.
14. Menambahkan Cek suhu pada Condensor Approach ($\leq 4.0^{\circ}\text{C}$) pada poin 7.3.18.
Add Check temperature on Condenser Approach ($\leq 4.0^{\circ}\text{C}$) on point 7.3.18.
15. Menambahkan Cek suhu pada Condensor Pressure (≤ 1000 kPa) pada poin 7.3.19.
Add Check the temperature at Condenser Pressure (≤ 1000 kPa) on point 7.3.19.
16. Menambahkan Cek tekanan pada Refrigerant Evaporator (250 kPa - 400 kPa) pada poin 7.3.20.
Add Check the pressure on the Refrigerant Evaporator (250 kPa - 400 kPa) on point 7.3.20.

17. Menambahkan Cek Calc. Evaporator Saturated Temperature (6°C - 7°C) pada poin 7.3.21.
add Check Calc. Evaporator Saturated Temperature (6°C - 7°C) on point 7.3.21.
18. Menambahkan Cek Evaporator Flow Delta Pressure Cut Out / Min 33.1 kPa (0.33 Bar) pada poin 7.3.22.
add Check Evaporator Flow Delta Pressure Cut Out / Min 33.1 kPa (0.33 Bar) on point 7.3.22.
19. Menambahkan Cek Condensor Flow Delta Pressure Cut Out / Min 34.5 kPa (0.34 Bar) pada poin 7.3.23.
add Check Condenser Flow Delta Pressure Cut Out / Mini 34.5 kPa (0.34 Bar) on point 7.3.23.
20. Menambahkan Cek Chiller Watter Pressure IN (3 - 5 Bar) pada poin 7.3.24.
add Check Chilled Water Pressure IN (3 - 5 Bar) on point 7.3.24.
21. Menambahkan Cek Chiller Watter Pressure OUT (2.5 - 4.5 Bar) pada poin 7.3.25.

add Check Chiller Watter Pressure OUT (2.5 - 4.5 Bar) on point 7.3.25.

22. Menambahkan Cek Condensor watter pressure IN (≥ 1.5 Bar) pada poin 7.3.26.
add Check Condenser water pressure IN (≥ 1.5 Bar) on point 7.3.26.

23. Menambahkan Cek Condensor watter pressure OUT (≥ 1 Bar) pada poin 7.3.27.
add Check Condenser water pressure OUT (≥ 1 Bar) on point 7.3.27.

24. Menambahkan Cek Refrigerant Pressure/Pressure Gauge (100 psi - 145 psi) pada poin 7.3.28.
add Check Refrigerant Pressure/Pressure Gauge (100 psi - 145 psi) on point 7.3.28.

25. Menambahkan Cek Rectifier Temperature ($\leq 71^{\circ}\text{C}$) pada poin 7.3.29.
add Check Rectifier Temperature ($\leq 71^{\circ}\text{C}$) on point 7.3.29.

26. Menambahkan Cek Inverter Temperature ($\leq 71^{\circ}\text{C}$) pada poin 7.3.30.

Added Inverter Temperature Check ($\leq 71^{\circ}\text{C}$) on point 7.3.30.

27. Merevisi poin 7.6.1.2 menjadi
Cek kondisi prefilter G4 ganti jika sudah kotor. Jika AHU dilengkapi dengan precooling dan dehumidifier maka bersihkan filternya, pre filter (fresh air) diganti seminggu 1x setiap hari senin, diganti dengan filter yang sudah di cuci (maksimal pencucian filter yg boleh dipakai adalah 10x, setelah di cuci 10x maka filter harus diganti yang baru).
Revise point 7.6.1.2. Check the condition of the G4 prefilter, replace it if it is dirty. If the AHU is equipped with precooling and dehumidifier then clean the filter, replace the pre filter (fresh air) once a week every Monday, replace it with a filter that has been washed (the maximum filter wash that can be used is 10x, after washing it 10x then the filter must be replaced. new).
28. Menambahkan Penggantian HEPA Filter H14 dilakukan secara berkala setiap 3 tahun sekali atau jika differential pressure masih memenuhi syarat yaitu 90 – 500 Pascal

artinya filter masih dapat digunakan pada poin 7.6.3.7.
Replacement of HEPA filter H14 is done periodically every 3 years or if the differential pressure is still eligible 90 – 500 Pascal means the filter can still be used on pont 7.6.3.7.

29. Menambahkan pergantian filter H14 HEPA filter di bagian (L4) pada formulir SOP/EBI/EN-016-F08F
Added H14 HEPA filter replacement in section (L4) on the SOP/EBI/EN-016-F08F form

30. Merevisi Formulir SOP/EBI/EN-016-F01G dengan mengganti kolom inspection menjadi number of cleaning dan menghapus keterangan change & no change pada kolom cleaning dikarenakan sudah tidak relevan.
Revise the SOP/EBI/EN-016-F01G Form by changing the inspection column to number of cleaning and deleting the change & no change information in the cleaning column because it is no longer relevant

			31. add Check Inverter Temperature ($\leq 71^{\circ}\text{C}$) on point 7.3.30. 32. Menambahkan cek kondisi insulasi pipa drain AHU pada poin 7.6.1.3. <i>Added check of AHU drain pipe insulation condition on point 7.6.1.3.</i> 33. Menambahkan check drain insulation AHU pada formulir SOP/EBI/EN-016-F08G <i>Add AHU drain insulation check to the SOP/EBI/EN-016-F08G form</i>
	CC-EBI-QA-051-24	07	1. Menambahkan keterangan pada poin 5.10. Disarankan untuk melakukan pengkondisian HVAC selama 7 jam agar parameter suhu dan RH memenuhi spesifikasi dan memerlukan 35 jam untuk pengkondisian HVAC setelah perawatan AHU jika unit AHU yang mati dan boiler. Tim produksi perlu memastikan bahwa semua parameter (suhu, kelembaban relatif, dan perbedaan tekanan) memenuhi spesifikasi dan melakukan pembersihan ruangan sesuai dengan SOP/EBI/PD-027 versi

terbaru sebelum proses produksi.

Add information to point 5.10. It is suggested for conditioning HVAC 7 hours for the temperature and RH parameters to meet the required specifications and need 35 hours for conditioning HVAC after maintenance AHU if AHU unit off and boiler off. Production team need to ensure that all parameters (temperature, relative humidity and differential pressure) meet specification and perform line clearance according to SOP/EBI/PD-027 latest version before production process.

2. Merevisi poin 7.6.3.1 menjadi Cek filter medium (F7) rigid, bag filter dan H10 hepa filter, pergantian dilakukan jika differential pressure sudah mencapai action limit atau kotor (blocking)

Revise point 7.6.3.1 to Check the rigid medium filter (F7), bag filter and H10 hepa filter, replacement is carried out if the differential pressure has reached the action limit or is dirty (blocking)

3. Merevisi poin 7.6.3.7. menjadi Cek HEPA Filter H14 dilakukan secara berkala, penggantian setiap 3 tahun sekali atau jika differential pressure masih memenuhi syarat yaitu 90 – 500 Pascal artinya filter masih dapat digunakan.

Revise point 7.6.3.7. to Check HEPA Filter H14 periodically, replace every 3 years or if the differential pressure still meets the requirements, namely 90 - 500 Pascal, meaning the filter can still be used.

4. Merevisi poin 7.9.1. Cek filter karbon atau bag filter F6 dan filter bibo, pergantian dilakukan jika differential pressure sudah mencaapai action limit atau kotor (blocking).

Revise point 7.9.1. Check the carbon filter or bag filter F6 and bibo filter, replacement is carried out if the differential pressure has reached the action limit or is dirty (blocking).

5. Menghilangkan AHU B106 pada kolom F7 BAG di formulir SOP/EBI/EN-016-

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	SOP No.	: SOP/EBI/EN-016
	Revision	: 07

			F03G dikarenakan sudah tidak relevan. <i>Remove AHU B106 in column F7 BAG in the SOP/EBI/EN-016-F03G form because it is no longer relevant.</i>
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11. DISTRIBUSI

DISTRIBUTION

11.1. Departemen Pemastian Mutu

Quality Assurance Department

11.2. Departemen Engineering

Engineering Department

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK		Document No. : SOP/EBI/EN-016-F01G
	PEMANTAUAN HARIAN TERHADAP SISTEM HVAC, CHILLER & HOT WATER		Effective Date :
	DAILY MONITORING OF HVAC SYSTEM, CHILLER & HOT WATER		

Date :

Checklist Items	Standard Value	Chiller No.	Time : (Shift 1, Shift 2 & Shift 3)			Remark
			09:00	17:00	02:00	
LCL Set Point	6°C - 7°C					
ECL Set Point	9°C - 12°C					
% AMP IN	≥ 40 %					
Oil Level	≥ 50% (Sight Glass)					
Comp Motor Winding Temperature	≤ 93°C					
Comp Discharge Temperature	≤ 60°C					
Oil Pressure ΔP	Running Min 170 kPa					
Oil Sump Temp	≥ 30°C					
Condensor Temperature Liquid IN (CDL IN)	≥ 28°C					
Condensor Temperature Liquid OUT (CDL OUT)	30°C - 38°C					
Evaporator Temperature Liquid IN (CHL IN)	9°C - 12°C					
Evaporator Temperature Liquid OUT (CHL OUT)	6°C - 7°C					
Evap Approach	≤ 2.8°C					
Cond Approach	≤ 4.0°C					
Condenser Pressure	≤ 1000 kPa					
Evap Refrigerant Pressure (Running Normal)	250 kPa - 400 kPa					
Calc. Evap Saturated Temperature	6°C - 7°C					
Evap Flow Delta Pressure Cut Out	Min 33.1 kPa (0.33 Bar)					
Cond Flow Delta Pressure Cut Out	Min 34.5 kPa (0.34 Bar)					
Chiller Watter Pressure IN	3 - 5 Bar					
Chiller Watter Pressure OUT	2.5 - 4.5 Bar					
Condensor watter pressure IN	≥ 1.5 Bar					
Condensor watter pressure OUT	≥ 1 Bar					
Refrigerant Pressure (Pressure Gauge)	100 psi - 145 psi					
Rectifier Temperature	≤ 71°C					
Inverter Temperature	≤ 71°C					
Hot Water Return Temperature	20 - 55 °C					
CHW S – R Temperature ± 4°C	7 - 12°C					
CW S – R Temperature ± 4°C	31- 36°C					
Clean the room and chiller machine	Clean					
Note :	Operator :			Checked by :		
	Shift 1	Shift 2	Shift 3			
	()	()	()	()		

 PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / FORM		Document No. : SOP/EBI/EN-016-F06H
	PEMANTAUAN MINGGUAN TERHADAP FILTER HEPA H14		Effective Date :
	WEEKLY MONITORING OF HEPA FILTER H14		

Date: Every : Monday

No	Hepa	Time 10:30	No	Hepa	Time 10:30	No	Hepa	Time 17:30
1	A/L 1-18		38	Washing & Ster. Room 1-56 (4)		75	Gowning - Off C (1-75)	
2	Janitor 1-31		39	Bag Removal 1-57 (1)		76	Gowning - Off C (1-77)	
3	MAL 1 1-40		40	Bag Removal 1-57 (2)		77	A/L (1-78)	
4	Gowning D 1-41		41	Bag Removal 1-57 (3)		78	Personal Air Lock C (1-79)	
5	Laundry 1-43 (1)		42	Bag Removal 1-57 (4)		79	A/L (1-80)	
6	Laundry 1-43 (2)		43	Stagging before Weighing 1-58 (1)		80	Tools & Filter Room (1-51)	
7	Laundry 1-43 (3)		44	Stagging before Weighing 1-58 (2)		81	Mixing Buffer (1-55)	
8	Material Air Lock 1-44		45	Weighing 1-59		82	Filling Room 2 (1-29)	
9	A/L (1-45)		46	Formulation 1-60 (1)		83	Change Part Room (1-24)	
10	Gowning D (1-82)		47	Formulation 1-60 (2)		84	Formulation 2 (1-39)	
11	A/L (1-73)		48	Collecting Area 1-61 (1)		85	Janitor 2 WH-28	
12	Grade-C Corridor 1-46 (1)		49	Collecting Area 1-61 (2)		86	Sampling Material WH-29 (1)	
13	Grade-C Corridor 1-46 (2)		50	Collecting Area 1-61 (3)		87	Sampling Material WH-29 (2)	
14	Grade-C Corridor 1-46 (3)		51	Deter. Prep. & Sanitation 1-62 (1)		88	MAL WH-30	
15	Grade-C Corridor 1-46 (4)		52	Deter. Prep. & Sanitation 1-62 (2)		89	Washing WH-31	
16	Grade-C Corridor 1-46 (5)		53	Corridor 1-63 (1)		90	A/L WH-32	
17	Grade-C Corridor 1-46 (6)		54	Corridor 1-63 (2)		91	Gowning ON WH-33	
18	Grade-C Corridor 1-46 (7)		55	A/L 1-64 (1)		92	A/L WH-34	
19	Grade-C Corridor 1-46 (8)		56	A/L 1-64 (2)		93	Gowning Off WH-35	
20	Grade-C Corridor 1-46 (9)		57	Filling Room 1-65 (1)		94	Air Lock D Off Q-35	
21	Grade-C Corridor 1-46 (10)		58	Filling Room 1-65 (2)		95	Gowning D off Q-36 (1)	
22	Grade-C Corridor 1-46 (11)		59	Filling Room 1-65 (3)		96	Gowning D off Q-36 (2)	
23	Grade-C Corridor 1-46 (12)		60	Filling Room 1-65 (4)		97	A/L On Q-37	
24	Grade-C Corridor 1-46 (13)		61	Filling Room 1-65 (5)		98	Gowning C off Q-38	
25	Grade-C Corridor 1-46 (14)		62	Filling Room 1-65 (6)		99	A/L C on Q-39	
26	Control 1-47 (1)		63	Grade B Corridor 1-66 (1)		100	Limit Test Q-40 (1)	
27	Control 1-47 (2)		64	Grade B Corridor 1-66 (2)		101	Limit Test Q-40 (2)	
28	Waste Out 1-48		65	Grade B Corridor 1-66 (3)		102	Positve Control Q-41 (1)	
29	MAL 2 1-49		66	Grade B Corridor 1-66 (4)		103	Positve Control Q-41 (2)	
30	A/L 1-50		67	Waste & Equip. For Cleaning 1-67		104	Sterility Test Room Q-42 (1)	
31	Tools & Filter Room 1-51		68	Air Lock 1-68		105	Sterility Test Room Q-42 (2)	
32	Pre Filled Syringes 1-52		69	A/L (1-74)		106	Sterility Test Room Q-42 (3)	
33	Remnant Storage 1-53		70	Gowning ON 1-69		107	Corridor C Q-51 (1)	
34	Clean Equipment Storage 1-54		71	Gowning OFF 1-70		108	Corridor C Q-51 (2)	
35	Washing & Ster. Room 1-56 (1)		72	A/L 1-71		109	Preincubation Room (Q-29)	
36	Washing & Ster. Room 1-56 (2)		73	A/L 1-72		110	Clean Room Janitor (Q-53)	
37	Washing & Ster. Room 1-56 (3)		74	Gowning - On C (1-76)		111	Gowning C Off (Q-34)	

Acceptance Criteria: 90 - 500 Pa
Alert Limit: 100 Pa (Bocor), 485 Pa (Mampet)
Action Limit: 90 Pa (Bocor), 500 Pa (Mampet)

Note :	Operator : Shift 1 ()	Operator : Shift 2 ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / FORM		Document No. : SOP/EBI/EN-016-F06H
	PEMANTAUAN MINGGUAN TERHADAP FILTER HEPA H14		Effective Date :
	WEEKLY MONITORING OF HEPA FILTER H14		

Date:

No	Hepa	Time 04:00	No	Hepa	Time 04:00
1	Transfer Room 1 (2-6) 1		38	Clean Storage (2-12) 2	
2	Transfer Room 1 (2-6) 2		39	Clean Storage (2-12) 3	
3	Encapsulation (2-7) 1		40	Air Lock (2-13) 1	
4	Encapsulation (2-7) 2		41	Air Lock (2-13) 2	
5	Encapsulation (2-7) 3		42	Air Lock (2-13) 3	
6	Encapsulation (2-7) 4		43	Janitor (2-14)	
7	Encapsulation (2-7) 5		44	Receive (2-15) 1	
8	Encapsulation (2-7) 6		45	Receive (2-15) 2	
9	Storage Room (2-7a) 1		46	Air Lock (2-36)	
10	Mix Encapsulation (2-7a) 2		47	Laundry (2-48) 1	
11	Transfer Room 2 (2-8) 1		48	Laundry (2-48) 2	
12	Transfer Room 2 (2-8) 2		49	Laundry (2-48) 3	
13	Purification (2-9) 1		50	Change Room (2-50)	
14	Purification (2-9) 2		51	Debagging (1st cover) (2-51)	
15	Purification (2-9) 3		52	Air Lock (2-52)	
16	Purification (2-9) 4		53	MAL (2-53)	
17	Purification (2-9) 5		54	Gowning C (2-54)	
18	Purification (2-9) 6		55	Air Lock (2-67)	
19	Purification (2-9) 7		56	Gowning D (2-68)	
20	Purification (2-9) 8		57	Air Lock (2-71)	
21	Storage 2 Grade C (2-2) 1		58	MAL (2-59)	
22	Storage 2 Grade C (2-2) 2		59	Second Washer (2-60) 1	
23	Waste & Equipment for Cleaning (2-3) 1		60	Second Washer (2-60) 2	
24	Crade C Corridor (2-4) 1		61	Second Washer (2-60) 3	
25	Crade C Corridor (2-4) 2		62	Second Washer (2-60) 4	
26	Crade C Corridor (2-4) 3		63	Final Sterilization (2-61) 1	
27	Crade C Corridor (2-4) 4		64	Final Sterilization (2-61) 2	
28	Crade C Corridor (2-4) 5		65	Final Sterilization (2-61) 3	
29	Crade C Corridor (2-4) 6		66	Gowning ON (2-62)	
30	IVT Capping (2-5) 1		67	Air Lock (2-63)	
31	IVT Capping (2-5) 2		68	Air Lock (2-64)	
32	Weighing Room (2-5a) 1		69	Storage 1 Grade C (2-1) 1	
33	Weighing Room (2-5a) 2		70	Storage 1 Grade C (2-1) 2	
34	Buffer Preparation & Storage (2-10) 1		71	Storage 1 Grade C (2-1) 3	
35	Buffer Preparation & Storage (2-10) 2		72	Air Lock (2-47)	
36	Air Lock (2-11)				
37	Clean Storage (2-12) 1				
Acceptance Criteria: 90 - 500 Pa Alert Limit: 100 Pa (Bocor), 485 Pa (Mampet) Action Limit: 90 Pa (Bocor), 500 Pa (Mampet)					
Note :			Operator : Shift 3 ()		Checked by : ()

 PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / FORM		Document No. : SOP/EBI/EN-016-F07H
	PEMBERSIHAN FRESH AIR FILTER DAN PRE FILTER G4 AHU		Effective Date :
	CLEANING OF AHU FRESH AIR FILTER AND PRE FILTER G4		

Date:

No.	AHU Name	Every Monday (16:00)	
		Fresh Air Washabel (G4 Filter)	
		Number Of Cleaning	Change
1	B101		
2	B102		
3	B103		
4	B104		
5	B105		
6	F101		
7	F102		
8	F103		
9	F104		
10	F105		
11	F106		
No.	AHU Name	Every Monday (23:30)	
		Fresh Air Washabel (G4 Filter)	
		Number Of Cleaning	Change
12	A101		
13	A102		
14	B201		
15	B202		
16	B204		
17	B205		
18	B207A		
19	B207B		
20	B208		

Note :

cek kondisi prefilter G4 ganti jika sudah kotor. Jika AHU dilengkapi dengan precooling dan dehumidifier maka bersihkan filternya, pre filter (fresh air) diganti seminggu 1x setiap hari senin, diganti dengan filter yang sudah di cuci (maksimal pencucian filter yg boleh dipakai adalah 10x, setelah di cuci 10x maka filter harus diganti yang baru).

check the condition of the G4 prefilter, replace it if it is dirty. If the AHU is equipped with precooling and dehumidifier then clean the filter, replace the pre filter (fresh air) once a week every Monday, replace it with a filter that has been washed (the maximum filter wash that can be used is 10x, after washing it 10x then the filter must be replaced. new).

Note :	Operator :		Checked by :
	Shift 2	Shift 3	
	()	()	

 PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / FORM		Document No. : SOP/EBI/EN-016-F08H
	PERAWATAN BERKALA TERHADAP UNIT PENANGANAN UDARA		Effective Date :
	PERIODIC MAINTENANCE OF AIR HANDLING UNIT (AHU)		

Level	:										
Month	:										
Date	:										
Day	:										
AHU	:										
Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L1											
	Cek kondisi basin AHU										
	Prefilter G4										
	Prefilter ahu precooling										
	Filter of process and reactivation air dehumidifier										
	Cek insulasi drain AHU										
L2											
	Cooling coil and heating coil										
	Condensation basin & drainpipe										
	Blower motor fan										
	Blower motor fan dehumidifier										
	V-Belt motor										
	Rotor drive motor assembly										
	Dehumidification rotor										
	Check Electric current, cleaning electrical panel and filter panel										
L4											
	Check medium filter F7 bag filter ahu										
	Check medium filter F7 bag filter precooling										
	Check medium filter F7 rigid filter										
	Change medium filter H10 HEPA filter										
	Check filter H14 HEPA Filter										
	Condensation reservoir & drainpipe										
	Measure the isolation resistance and grounding, maximum 5Ω										
	All cable terminal connections										
	Connection from leaks										
	Inspect all circuit breakers										

Note :

No	Date	Start	Finish	Duration		Total personnel	Area	Remark
				Hour	Minute			

 PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / FORM		Document No. : SOP/EBI/EN-016-F08H
	PERAWATAN BERKALA TERHADAP UNIT PENANGANAN UDARA		Effective Date :
	<i>PERIODIC MAINTENANCE OF AIR HANDLING UNIT (AHU)</i>		

Remark : C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek)	Level : L1 : Every 1 month L2 : Every 3 month L4 : Every 1 year	Done by : ()	Checked by : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / FORM								Document No. : SOP/EBI/EN-016-F09H	
	PERAWATAN BERKALA TERHADAP UNIT KUMPARAN KIPAS									
	PERIODIC MAINTENANCE OF FAN COIL UNIT (FCU)								Effective Date :	

Level :											
Month :											
Date :											
Day :											
FCU :											
Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L1											
	Pre Filter G4										
	Cooling coil										
L2											
	Cooling coil										
	Condensation basin & drain pipe										
	Fan motor										
	Connection of the electrical										
	Electric current										
Note :											
No	Date	Start	Finish	Duration		Total personnel	Area	Remark			
				Hour	Minute						
Remark : C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek)			Level : L1 : Every 1 month L2 : Every 3 month		Done by : ()		Checked by : ()				

 PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / FORM								Document No. : SOP/EBI/EN-016-F10H	
	PERAWATAN BERKALA TERHADAP FAN EXHAUST									
	PERIODIC MAINTENANCE OF EXHAUST FAN								Effective Date :	

Level	:
Month	:
Date	:
Day	:
Exhaust Fan	:

Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L2											
	Blower motor fan										
	Electric current										
	V-Belt										
	Carbon filter										
	Bag filter F6										
L4											
	Change carbon filter										
	Change Bibo Filter										
	Change bag filter F6										
	Resistance isolation & grounding										
	Inspect all circuit bracker (CB)										

Note :

No	Date	Start	Finish	Duration		Total personnel	Area	Remark
				Hour	Minute			

Remark :

C : Cleaning (pembersihan)
 I : Inspection (pemeriksaan)
 L : Lubrication (pelumasan)
 R : Replacement (penggantian)
 S : Setting (penyetelan)
 T : Tightened (dikencangkan)
 W : Welding (pengelasan)
 G : Good (baik)
 B : Bad (jelek)

Level :

L2 : Every 3 month
 L4 : Every 1 year

Done by :

()

Checked by :

()

FORMULIR / FORM

PERAWATAN BERKALA TERHADAP CHILLER CARRIER
EVERGREEN 23XRV

PERIODIC MAINTENANCE OF CHILLER CARRIER
EVERGREEN 23XRV

Document No. :
SOP/EBI/EN-016-F11H

Effective Date :

Level :											
Month :											
Date :											
Day :											
Hours Meter :											
Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L2											
	Oil level										Normal 50%
	Control panel from dust										
	Pumps from leaks										
	Connection of the electrical										
	Electric current of the pump										
	Clean the strainer										
L4											
	Change compressor oil										
	Change oil filter										
	Change motor cooling filter dryer										
	Inspect & clean cooler tubes										
	Inspect function cooler relief valves										
	Change refrigerant filter dryer										
	Inspect & clean condenser tubes										
	Inspect function condenser relief valves										
	Change VFD refrigerant strainer										
	Resistance isolation & grounding										Max. 5 Ω
	Connection of the electrical										
	Mechanical seal pump										
	Inspect all circuit bracker (CB)										
Note :											
No	Date	Start	Finish	Duration		Total personnel	Area	Remark			
				Hour	Minute						
Remark :			Level :			Done by :		Checked by :			
C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek)			L2 : Every 3 month L4 : Every 1 year			()		()			

Level :											
Month :											
Date :											
Day :											
Cooling Tower No. :											
Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L2											
	Connections pipe and valves										
	Fan blower										
	Fins and water seservoir										
	Pump from leaks										
	Connection of the electrical										
	Electric current of the pump										
	Clean the strainer										
L4											
	Resistance isolation & grounding										Max. 5 Ω
	Connection of the electrical										
	Mechanical pump seal										
	Inspect all circuit bracker (CB)										
Note :											
No	Date	Start	Finish	Duration		Total personnel	Area	Remark			
				Hour	Minute						
Remark :			Level :			Done by :		Checked by :			
C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek)			L2 : Every 3 month L4 : Every 1 year			()		()			

Level :											
Month :											
Date :											
Day :											
Hours Meter :											
Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L2											
	Control panel from dust										
	Pump from leaks										
	Connection of the electrical										
	Electric current of the pump										
	Function of the safety valve										
	Clean the strainer										
L4											
	Steam traps from leaks and other										
	Mechanical seal pump										
	Cek and clean the PRV										
Note :											
No	Date	Start	Finish	Duration		Total personnel	Area	Remark			
				Hour	Minute						
Remark :				Level :		Done by :		Checked by :			
C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek)				L2 : Every 3 month L4 : Every 1 year		()		()			